

Pinaki Mazumder

professor

基本信息

姓名: Pinaki Mazumder

所属院校: 密歇根大学-安娜堡

地区: United States

学科领域

电子科学与技术

微电子科学与工程

应用物理学

集成电路设计与集成系统

电磁场与无线技术

物理学

通信工程

光电信息科学与工程

研究领域

太赫兹技术与应用

微波器件与电路

微纳电子与纳米技术

光电子器件与微纳光学

高频与微波技术

电磁场理论与数值计算

微电子器件与制造

电磁学与电动力学

纳米电子与先进工艺

光电子技术 with 光学工程

+10

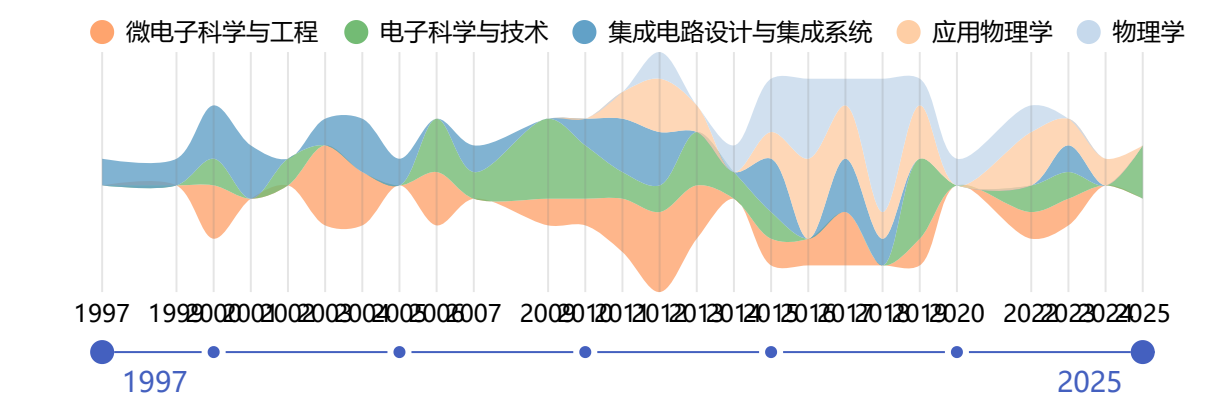
个人简介

Pinaki Mazumder is a Professor of Computer Science and Engineering at the University of Michigan. His research focuses on VLSI, nanoelectronics circuits, electronic design automation, terahertz technology, and emerging technologies. He has authored numerous publications on memristor devices, neuromorphic systems, resonant tunneling devices, and genetic algorithms for VLSI design. Mazumder's work significantly contributes to the advancement of digital circuit applications and nanotechnology.

工作经历

University of Michigan - Professor

近年发表的论文



Sub-terahertz transmissive reconfigurable intelligent surface for integrated beam steering and self-OOK-modulation

LIGHT-SCIENCE & APPLICATIONS

Shen, Dongfang; Lan, Feng; Wang, Luyang; Song, Tianyang; Yang, Munan; Hu, Tianyu; Li, Yueting; Nie, Xiaolei; Yang, Jiayao; Liang, Shixiong; Zeng, Hongxin; Zhang, Hui-Fang; Mazumder, Pinaki; Yang, Ziqiang; Zhang, Yaxin; Cui, Tie Jun

被引用次数: 0 2025

Programmable beam steering and quasi-continuous phase shift on a 2-bit terahertz coding metasurface with HEMT-switched multimodal modulation

OPTICS EXPRESS

Song, Tianyang; Lan, Feng; Wang, Luyang; Zeng, Hongxin; Liang, Shixiong; Yang, Munan; Shen, Dongfang; Liu, Wenxin; Mazumder, Pinaki; Zhang, Yaxin; Yang, Ziqiang

被引用次数: 0 2025

Terahertz Coding Metasurface with Low-Switch-Ratio for Rapid 1-bit Phase Modulation and Beam Steering

ACS PHOTONICS

Song, Tianyang; Lan, Feng; Wang, Luyang; Zeng, Hongxin; Liang, Shixiong; Yang, Munan; Shen, Dongfang; Liu, Wenxin; Jin, Shengxiao; Zhu, Zhongbo; Mazumder, Pinaki; Zhang, Yaxin; Yang, Ziqiang

被引用次数: 0 2025

Scalable almost-linear dynamical Ising machines

NATURAL COMPUTING

Shukla, Aditya; Erementchouk, Mikhail; Mazumder, Pinaki

被引用次数: 0 2024

Enhanced terahertz diffuse scattering through convolutional coding on broadband polarization conversion metasurface

OPTICS EXPRESS

Song, Tianyang; Lan, Feng; Wang, Luyang; Yang, Munan; Zeng, Hongxin; Liu, Wenxin; Mazumder, Pinaki; Zhang, Yaxin; Yang, Ziqiang

被引用次数: 0 2024

Real-time programmable metasurface for terahertz multifunctional wave front engineering

LIGHT-SCIENCE & APPLICATIONS

Lan, Feng; Wang, Luyang; Zeng, Hongxin; Liang, Shixiong; Song, Tianyang; Liu, Wenxin; Mazumder, Pinaki; Yang, Ziqiang; Zhang, Yaxin; Mittleman, Daniel M.

被引用次数: 26

2023

Collective-Coupling Enhanced Ultrabroadband Linear Polarization Conversion on Zigzag-Split Metasurfaces

IEEE TRANSACTIONS ON ANTENNAS AND PROPAGATION

Yang, Munan; Lan, Feng; Zhang, Yaxin; Qi, Limei; He, Guiju; Pan, Yibo; Song, Tianyang; Wang, Luyang; Mazumder, Pinaki; Zeng, Hongxin; Yang, Ziqiang

被引用次数: 2

2023

Custom CMOS Ising Machine Based on Relaxed Burer-Monteiro-Zhang Heuristic

IEEE TRANSACTIONS ON COMPUTERS

Shukla, Aditya; Erementchouk, Mikhail; Mazumder, Pinaki

被引用次数: 0

2023

Broadband Nonuniform Terahertz Multimode Conversion Series with Compactness and Pure Pattern

JOURNAL OF INFRARED MILLIMETER AND TERAHERTZ WAVES

Lan, Feng; He, Guiju; Mazumder, Pinaki; Yan, Yang; Meng, Lin; Yang, Ziqiang; Zhang, Yaxin; Shi, Zongjun; Liu, Wenxin; Song, Tianyang; Wang, Luyang

被引用次数: 0

2022

Dynamic Pinning Synchronization of Fuzzy-Dependent-Switched Coupled Memristive Neural Networks With Mismatched Dimensions on Time Scales

IEEE TRANSACTIONS ON FUZZY SYSTEMS

Wang, Xiangxiang; Yu, Yongbin; Cai, Jingye; Zhong, Shouming; Yang, Nijing; Shi, Kaibo; Mazumder, Pinaki; Tashi, Nyima

被引用次数: 13

2022

Bayesian Depth-Wise Convolutional Neural Network Design for Brain Tumor MRI Classification

DIAGNOSTICS

Ekong, Favour; Yu, Yongbin; Patamia, Rutherford Agbeshi; Feng, Xiao; Tang, Qian; Mazumder, Pinaki; Cai, Jingye

被引用次数: 10

2022

On computational capabilities of Ising machines based on nonlinear oscillators

PHYSICA D-NONLINEAR PHENOMENA

Erementchouk, Mikhail; Shukla, Aditya; Mazumder, Pinaki

被引用次数: 6

2022

Ultra-broadband Terahertz Complex Amplitude Modulation by HEMT-Switch Control 1-bit Coding Microstructure

2022 CROSS STRAIT RADIO SCIENCE & WIRELESS TECHNOLOGY CONFERENCE, CSRSWTC

Xiao, Zong; Lan, Feng; Yang, Ziqiang; Wang, Luyang; Song, Tianyang; Wang, Shun; Wang, Yujian; Mazumder, Pinaki

被引用次数: 0

2022

Navigating the Seismic Shift of Post-Moore Computer Systems Design

IEEE MICRO

Banerjee, Anindya; Basu, Sankar; Brunvand, Erik; Mazumder, Pinaki; Cleaveland, Branka; Singh, Gurdip; Martonosi, Margaret; Pembleton, Fernanda

被引用次数: 1

2021

A fractional phase-coding strategy for terahertz beam patterning on digital metasurfaces

OPTICS EXPRESS

Wang, Luyang; Lan, Feng; Zhang, Yaxin; Liang, Shixiong; Liu, Wenxin; Yang, Ziqiang; Meng, Lin; Shi, Zongjun; Yin, Jing; Song, Tianyang; Zeng, Hongxin; Mazumder, Pinaki

被引用次数: 19

2020

Continuous-variable quantum key distribution with discretized modulations in the strong noise regime

PHYSICAL REVIEW A

Erementchouk, Mikhail; Mazumder, Pinaki

被引用次数: 3

2020

Broadband terahertz reconfigurable metasurface based on 1-bit asymmetric coding metamaterial

OPTICS COMMUNICATIONS

Zeng, Hongxin; Lan, Feng; Zhang, Yaxin; Liang, Shixiong; Wang, Lan; Yin, Jing; Song, Tianyang; Wang, Luyang; Zhang, Ting; Shi, Zongjun; Yang, Ziqiang; Mazumder, Pinaki

被引用次数: 16

2020

Terahertz Quantum Cryptography

IEEE JOURNAL ON SELECTED AREAS IN COMMUNICATIONS

Ottaviani, Carlo; Woolley, Matthew J.; Erementchouk, Misha; Federici, John F.; Mazumder, Pinaki; Pirandola, Stefano; Weedbrook, Christian

被引用次数: 32

2020

Evolution map of the memristor: from pure capacitive state to resistive switching state

NANOSCALE

Zhou, Guangdong; Wu, Jinggao; Wang, Lidan; Sun, Bai; Ren, Zhijun; Xu, Cunyun; Yao, Yanqing; Liao, Liping; Wang, Gang; Zheng, Shaohui; Mazumder, Pinaki; Duan, Shukai; Song, Qunliang

被引用次数: 48

2019

Large phase modulation of THz wave via an enhanced resonant active HEMT metasurface

NANOPHOTONICS

Zhang, Yaxin; Zhao, Yuncheng; Liang, Shixiong; Zhang, Bo; Wang, Lan; Zhou, Tianchi; Kou, Wei; Lan, Feng; Zeng, Hongxin; Han, Jiaguang; Feng, Zhihong; Chen, Qin; Mazumder, Pinaki; Yang, Ziqiang

被引用次数: 73

2019

Maximizing beam-scanning angle in an expected bandwidth based on terahertz metasurface with dual-mode resonance

APPLIED PHYSICS EXPRESS

Zeng, Hongxin; Lan, Feng; Zhang, Yaxin; Song, Tianyang; Liang, Shixiong; Wang, Lan; Zhang, Ting; Wang, Luyang; Shi, Zongjun; Zhang, Xilin; Mazumder, Pinaki; Yang, Ziqiang

被引用次数: 4

2019

Terahertz Dual-Polarization Beam Splitter Via an Anisotropic Matrix Metasurface

IEEE TRANSACTIONS ON TERAHERTZ SCIENCE AND TECHNOLOGY

Zeng, Hongxin; Zhang, Yaxin; Lan, Feng; Liang, Shixiong; Wang, Lan; Song, Tianyang; Zhang, Ting; Shi, Zongjun; Yang, Ziqiang; Kang, Xue; Zhang, Xilin; Mazumder, Pinaki; Mittleman, Daniel M.

被引用次数: 29

2019

Dual-band refractometric terahertz biosensing with intense wave-matter-overlap microfluidic channel

BIOMEDICAL OPTICS EXPRESS

Lan, Feng; Luo, Feng; Mazumder, Pinaki; Yang, Ziqiang; Meng, Lin; Bao, Zhengqiang; Zhou, Jun; Zhang, Yaxin; Liang, Shixiong; Shi, Zongjun; Khan, Abdur Rauf; Zhang, Ziqi; Wang, Luyang; Yin, Jing; Zeng, Hongxin

被引用次数: 61

2019

Spoof Plasmon Interconnects-Communications Beyond RC Limit

IEEE TRANSACTIONS ON COMMUNICATIONS

Joy, Soumitra Roy; Erementchouk, Mikhail; Yu, Hao; Mazumder, Pinaki

被引用次数: 30

2019

Properties of spoof plasmon in thin structures (vol 474, 20180205, 2018)

PROCEEDINGS OF THE ROYAL SOCIETY A-MATHEMATICAL PHYSICAL AND ENGINEERING SCIENCES

Joy, Soumitra Roy; Yu, Hao; Mazumder, Pinaki

被引用次数: 0

2019

Fault Modeling and Parallel Testing for 1T1M Memory Array

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Sun, Li; Zheng, Nan; Zhang, Tao; Mazumder, Pinaki

被引用次数: 15

2018

Enhanced quadruple-resonant terahertz metamaterial with asymmetric hybrid resonators

OPTICAL MATERIALS

Shi, Minglei; Lan, Feng; Mazumder, Pinaki; Aghadjani, Mandi; Yang, Ziqiang; Meng, Lin; Zhou, Jun

被引用次数: 5

2018

Properties of spoof plasmon in thin structures

PROCEEDINGS OF THE ROYAL SOCIETY A-MATHEMATICAL PHYSICAL AND ENGINEERING SCIENCES

Joy, Soumitra Roy; Yu, Hao; Mazumder, Pinaki

被引用次数: 6

2018

Weyl fermions in cylindrical wires

PHYSICAL REVIEW B

Erementchouk, Mikhail; Mazumder, Pinaki

被引用次数: 6

2018

Spatial accumulation of phase difference in spoof plasmon based Mach-Zehnder Interferometer

OPTICS COMMUNICATIONS

Aghadjani, Mahdi; Erementchouk, Mikhail; Mazumder, Pinaki

被引用次数: 2

2018

Force distribution inside the spoof surface plasmon polariton waveguide

JOURNAL OF THE OPTICAL SOCIETY OF AMERICA B-OPTICAL PHYSICS

Aghadjani, M.; Erementchouk, M.; Mazumder, P.

被引用次数: 3

2018

Online Supervised Learning for Hardware-Based Multilayer Spiking Neural Networks Through the Modulation of Weight-Dependent Spike-Timing-Dependent Plasticity

IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS

Zheng, Nan; Mazumder, Pinaki

被引用次数: 37

2018

A Scalable Low-Power Reconfigurable Accelerator for Action-Dependent Heuristic Dynamic Programming

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Zheng, Nan; Mazumder, Pinaki

被引用次数: 2

2018

Learning in Memristor Crossbar-Based Spiking Neural Networks Through Modulation of Weight-Dependent Spike-Timing-Dependent Plasticity

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Zheng, Nan; Mazumder, Pinaki

被引用次数: 46

2018

Spoof surface plasmon resonant tunneling mode with high quality and Purcell factors

PHYSICAL REVIEW B

Joy, Soumitra Roy; Erementchouk, Mikhail; Mazumder, Pinaki

被引用次数: 18

2017

Helicity-dependent scattering in layered Weyl semimetals

PHYSICS LETTERS A

Erementchouk, Mikhail; Mazumder, Pinaki

被引用次数: 2

2017

Ka-band relativistic diffraction generator with a tapered coaxial Bragg reflector

AIP ADVANCES

Lan, Feng; Yang, Ziqiang; Mazumder, Pinaki; Shi, Zongjun; Meng, Lin; Liu, Dagang; Fu, Wenjie

被引用次数: 2

2017

Hardware-Friendly Actor-Critic Reinforcement Learning Through Modulation of Spike-Timing-Dependent Plasticity

IEEE TRANSACTIONS ON COMPUTERS

Zheng, Nan; Mazumder, Pinaki

被引用次数: 16

2017

Modeling and Mitigation of Static Noise Margin Variation in Subthreshold SRAM Cells

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Zheng, Nan; Mazumder, Pinaki

被引用次数: 13

2017

Energy-Efficient Hardware Architecture of Self-Organizing Map for ECG Clustering in 65-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kim, Jaeyoung; Mazumder, Pinaki

被引用次数: 12

2017

An Efficient Eligible Error Locator Polynomial Searching Algorithm and Hardware Architecture for One-Pass Chase Decoding of BCH Codes

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Zheng, Nan; Mazumder, Pinaki

被引用次数: 1

2017

A robust 12T SRAM cell with improved write margin for ultra -low power applications in 40 nm CMOS

INTEGRATION-THE VLSI JOURNAL

Kim, Jaeyoung; Mazumder, Pinaki

被引用次数: 29

2017

A Drift-Tolerant Read/Write Scheme for Multilevel Memristor Memory

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Yilmaz, Yalcin; Mazumder, Pinaki

被引用次数: 23

2017

Towards developing a compact model for magnetization switching in straintronics magnetic random access memory devices

JOURNAL OF APPLIED PHYSICS

Barangi, Mahmood; Erementchouk, Mikhail; Mazumder, Pinaki

被引用次数: 7

2016

Dirac electrons in the presence of a matrix potential barrier: application to graphene and topological insulators

JOURNAL OF PHYSICS-CONDENSED MATTER

Erementchouk, Mikhail; Mazumder, Pinaki; Khan, M. A.; Leuenberger, Michael N.

被引用次数: 9	2016
Spoof Surface Plasmon Polariton Beam Splitter	
IEEE TRANSACTIONS ON TERAHERTZ SCIENCE AND TECHNOLOGY	
Aghadjani, Mahdi; Erementchouk, Mikhail; Mazumder, Pinaki	
被引用次数: 25	2016
Digital implementation of a virtual insect trained by spike-timing dependent plasticity	
INTEGRATION-THE VLSI JOURNAL	
Mazumder, P.; Hu, D.; Ebong, I.; Zhang, X.; Xu, Z.; Ferrari, S.	
被引用次数: 15	2016
Straintronics-Based True Random Number Generator for High-Speed and Energy-Limited Applications	
IEEE TRANSACTIONS ON MAGNETICS	
Barangi, Mahmood; Chang, Joseph S.; Mazumder, Pinaki	
被引用次数: 18	2016
Electrodynamics of spoof plasmons in periodically corrugated waveguides	
PROCEEDINGS OF THE ROYAL SOCIETY A-MATHEMATICAL PHYSICAL AND ENGINEERING SCIENCES	
Erementchouk, Mikhail; Joy, Soumitra Roy; Mazumder, Pinaki	
被引用次数: 31	2016
Memristor-Based Cellular Nonlinear/Neural Network: Design, Analysis, and Applications	
IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS	
Duan, Shukai; Hu, Xiaofang; Dong, Zhekang; Wang, Lidan; Mazumder, Pinaki	
被引用次数: 232	2015
Effect of temperature variations and thermal noise on the static and dynamic behavior of straintronics devices	
JOURNAL OF APPLIED PHYSICS	
Barangi, Mahmood; Mazumder, Pinaki	
被引用次数: 6	2015
THz Polarizer Controller Based on Cylindrical Spoof Surface Plasmon Polariton (C-SSPP)	
IEEE TRANSACTIONS ON TERAHERTZ SCIENCE AND TECHNOLOGY	
Aghadjani, Mahdi; Mazumder, Pinaki	
被引用次数: 14	2015
Terahertz Switch Based on Waveguide-Cavity-Waveguide Comprising Cylindrical Spoof Surface Plasmon Polariton	
IEEE TRANSACTIONS ON ELECTRON DEVICES	
Aghadjani, Mahdi; Mazumder, Pinaki	
被引用次数: 10	2015
Straintronics-Based Random Access Memory as Universal Data Storage Devices	
IEEE TRANSACTIONS ON MAGNETICS	

Barangi, Mahmood; Mazumder, Pinaki

被引用次数: 7

2015

Straintronics-based magnetic tunneling junction: Dynamic and static behavior analysis and material investigation

APPLIED PHYSICS LETTERS

Barangi, Mahmood; Mazumder, Pinaki

被引用次数: 20

2014

Terahertz Beam Steering With Doped GaAs Phase Modulator and a Design of Spatial-Resolved High-Speed Terahertz Analog-to-Digital Converter

IEEE TRANSACTIONS ON ELECTRON DEVICES

Xu, Zhao; Mazumder, Pinaki

被引用次数: 9

2014

Tunneling Through Finite Quantum Dot Superlattices

ARABIAN JOURNAL FOR SCIENCE AND ENGINEERING

Rajagopalan, M.; Mazumder, P.

被引用次数: 0

2014

Color Image Processing with Multi-Peak Resonant Tunneling Diodes

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Lee, Woo Hyung; Mazumder, Pinaki

被引用次数: 1

2013

Nonvolatile Nanopipelining Logic Using Multiferroic Single-Domain Nanomagnets

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Yilmaz, Yalcin; Mazumder, Pinaki

被引用次数: 14

2013

Image Processing by a Programmable Grid Comprising Quantum Dots and Memristors

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Yilmaz, Yalcin; Mazumder, Pinaki

被引用次数: 10

2013

Design of Highly Selective Metamaterials for Sensing Platforms

IEEE SENSORS JOURNAL

Song, Kyungjun; Mazumder, Pinaki

被引用次数: 11

2013

Nonlinear spoof surface plasmon polariton phenomena based on conductor metamaterials

PHOTONICS AND NANOSTRUCTURES-FUNDAMENTALS AND APPLICATIONS

Song, Kyungjun; Mazumder, Pinaki

被引用次数: 6

2012

Memristors: Devices, Models, and Applications

PROCEEDINGS OF THE IEEE

Mazumder, Pinaki; Kang, Sung Mo; Waser, Rainer

被引用次数: 160

2012

CMOS and Memristor-Based Neural Network Design for Position Detection

PROCEEDINGS OF THE IEEE

Ebong, Idongesit E.; Mazumder, Pinaki

被引用次数: 144

2012

Memristor-based RRAM with applications

SCIENCE CHINA-INFORMATION SCIENCES

Duan ShuKai; Hu XiaoFang; Wang LiDan; Li ChuanDong; Mazumder, Pinaki

被引用次数: 56

2012

Bio-Sensing by Mach-Zehnder Interferometer Comprising Doubly-Corrugated Spoofed Surface Plasmon Polariton (DC-SSPP) Waveguide

IEEE TRANSACTIONS ON TERAHERTZ SCIENCE AND TECHNOLOGY

Xu, Zhao; Mazumder, Pinaki

被引用次数: 29

2012

Analysis of Doubly Corrugated Spoof Surface Plasmon Polariton (DC-SSPP) Structure With Sub-Wavelength Transmission at THz Frequencies

IEEE TRANSACTIONS ON TERAHERTZ SCIENCE AND TECHNOLOGY

Xu, Zhao; Song, Kyungjun; Mazumder, Pinaki

被引用次数: 27

2012

Guest Editors' Introduction: Special Section on Chips and Architectures for Emerging Technologies and Applications

IEEE TRANSACTIONS ON COMPUTERS

Benso, Alfredo; Makris, Yiorgos; Mazumder, Pinaki

被引用次数: 0

2011

Dynamic Terahertz Spoof Surface Plasmon-Polariton Switch Based on Resonance and Absorption

IEEE TRANSACTIONS ON ELECTRON DEVICES

Song, Kyungjun; Mazumder, Pinaki

被引用次数: 36

2011

Self-Controlled Writing and Erasing in a Memristor Crossbar Memory

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Ebong, Idongesit E.; Mazumder, Pinaki

被引用次数: 62

2011

Equivalent-Circuit Modeling of Nonradiative Surface Plasmon Energy Transfer Along the Metallic Nanowire

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Song, Kyungjun; Mazumder, Pinaki

被引用次数: 1

2011

Nanoscale Memristor Device as Synapse in Neuromorphic Systems

NANO LETTERS

Jo, Sung Hyun; Chang, Ting; Ebong, Idongesit; Bhadviya, Bhavitavya B.; Mazumder, Pinaki; Lu, Wei

被引用次数: 3637

2010

One-Dimensional Surface Plasmon Photonic Crystal Slab (SPPCS) for a Nanophotodiode

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Song, Kyungjun; Mazumder, Pinaki

被引用次数: 0

2010

Tunneling-Based Cellular Nonlinear Network Architectures for Image Processing

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Mazumder, Pinaki; Li, Sing-Rong; Ebong, Idongesit E.

被引用次数: 38

2009

Active Terahertz Spoof Surface Plasmon Polariton Switch Comprising the Perfect Conductor Metamaterial

IEEE TRANSACTIONS ON ELECTRON DEVICES

Song, Kyungjun; Mazumder, Pinaki

被引用次数: 66

2009

An Equivalent Circuit Modeling of an Equispaced Metallic Nanoparticles (MNPs) Plasmon Wire

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Song, Kyungjun; Mazumder, Pinaki

被引用次数: 14

2009

Motion detection by quantum-dots-based velocity-tuned filter

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Lee, Woo Hyung; Mazumder, Pinaki

被引用次数: 9

2008

Accelerated chip-level thermal analysis using multilayer Green's function

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Wang, Baohua; Mazumder, Pinaki

被引用次数: 39

2007

Efficient Modeling of transmission lines with electromagnetic wave coupling by using the finite difference quadrature method

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Xu, Qinwei; Mazumder, Pinaki

被引用次数: 8

2007

Gate current modeling of high-k stack nanoscale MOSFETs

SOLID-STATE ELECTRONICS

Wang, Wei; Gu, Ning; Sun, J. P.; Mazumder, P.

被引用次数: 15

2006

Modeling of gate current and capacitance in nanoscale-MOS structures

IEEE TRANSACTIONS ON ELECTRON DEVICES

Sun, J. P.; Wang, Wei; Toyabe, Toru; Gu, Ning; Mazumder, Pinaki

被引用次数: 11

2006

Performance modeling of resonant tunneling-based random-access memories

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Zhang, H; Mazumder, P; Ding, L; Yang, KH

被引用次数: 6

2005

On circuit techniques to improve noise immunity of CMOS dynamic logic

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Ding, L; Mazumder, P

被引用次数: 74

2004

Noise-tolerant quantum MOS circuits using resonant tunneling devices

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Ding, L; Mazumder, P

被引用次数: 10

2004

Accurate crosstalk noise modeling for early signal integrity analysis

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Ding, L; Blaauw, D; Mazumder, P

被引用次数: 35

2003

Equivalent-circuit interconnect modeling based on the fifth-order differential quadrature methods

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Xu, QW; Mazumder, P

被引用次数: 10

2003

Simultaneous switching noise analysis using application specific device modeling

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Ding, L; Mazumder, P

被引用次数: 6

2003

Accurate modeling of lossy nonuniform transmission lines by using differential quadrature methods

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Xu, QW; Mazumder, P

被引用次数: 19

2002

Augmentation of SPICE for simulation of circuits containing resonant tunneling diodes

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Bhattacharya, M; Mazumder, P

被引用次数: 30

2001

On optimal tapering of FET chains in high-speed CMOS circuits

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Ding, L; Mazumder, P

被引用次数: 9

2001

A physical design tool for built-in self-repairable RAMs

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Chakraborty, K; Kulkarni, S; Bhattacharya, M; Mazumder, P; Gupta, A

被引用次数: 16

2001

Digital integrated circuit using integrated InAlAs/InGaAs/InP HEMTs and InAs/AlSb/GaSb RITDs

ELECTRONICS LETTERS

Fay, P; Lu, J; Xu, Y; Bernstein, GH; Gonzalez, A; Mazumder, P; Chow, DH; Schulman, JN

被引用次数: 6

2001

Time-domain modeling of high-speed interconnects by modified method of characteristics

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Xu, QW; Li, ZF; Mazumder, P; Mao, JF

被引用次数: 11

2000

Noise margins of threshold logic gates containing resonant tunneling diodes

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Bhattacharya, M; Mazumder, P

被引用次数: 5

2000

Redundant arithmetic, algorithms and implementations

INTEGRATION-THE VLSI JOURNAL

González, AF; Mazumder, P

被引用次数: 27

2000

New march tests for multiport RAM devices

JOURNAL OF ELECTRONIC TESTING-THEORY AND APPLICATIONS

Chakraborty, K; Mazumder, P

被引用次数: 3

2000

Special issue on quantum devices and their applications

PROCEEDINGS OF THE IEEE

Seabaugh, AC; Mazumder, P

被引用次数: 6

1999

Design and analysis of resonant-tunneling-diode (RTD) based high performance memory system

IEICE TRANSACTIONS ON ELECTRONICS

Uemura, T; Mazumder, P

被引用次数: 9

1999

Tunneling devices and applications in high functionality/speed digital circuits

SOLID-STATE ELECTRONICS

Haddad, GI; Mazumder, P

被引用次数: 35

1997

AN ON-CHIP ECC CIRCUIT FOR CORRECTING SOFT ERRORS IN DRAMS WITH TRENCH CAPACITORS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

MAZUMDER, P

被引用次数: 18

1992

Terahertz Quantum Cryptography

JSAC

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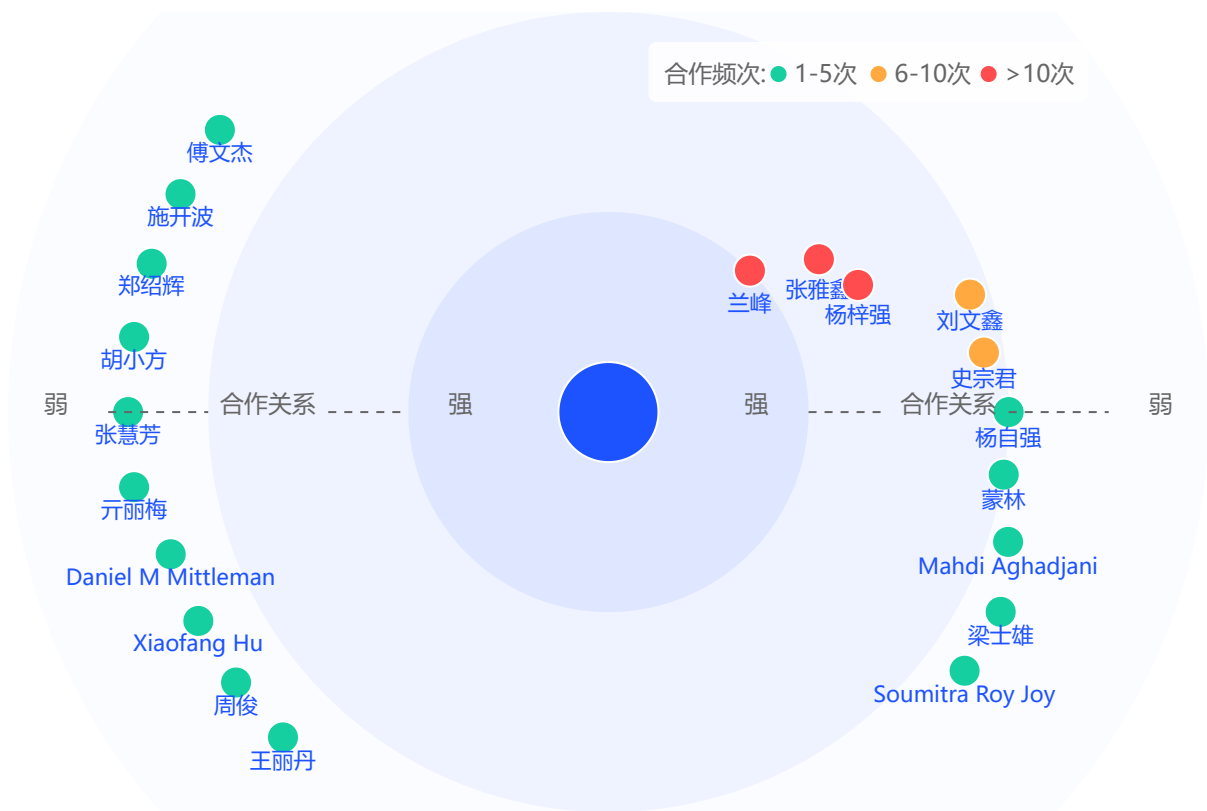
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